# Ram Reformulation of the Spacetime Penrose Composite

Domains, Exact Nonlinear Variations, Charge Periods,
Original-Data Rigidity and Local Schwarzschild–AdS Analysis

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**Abstract**

This report studies the pinned composite *Spacetime Penrose inequalities: enclosing area, charge, and rigidity*, including its neutral three- and four-dimensional spatial formulations, charged comparisons, original-data equality arguments, and fixed-seed local Schwarzschild–anti-de Sitter analysis. All 47 TeX source files were read in literal, recorded chunks; reading is not formal verification. The analysis retains the actual graph–conformal system, its state-dependent axial coefficients and inner conormal condition. We derive its complete joint differential and a boundary-correct weak Ram representation, prove a bordered local inverse at a simple zero principal MOTS eigenvalue, quantify metric sensitivity of the full enclosing-area infimum, identify exactly which electromagnetic periods the comparison discards, and prove a weighted TT projection identity with an explicit divergence-defect extension. The AdS discussion separates quadratic positivity, the four-dimensional nullspace, quartic positivity and exact radial equality, with seed-dependent remainder thresholds. Global existence, obstacle regularity, rest-frame reduction, positive mass, static classification and the composite's final global theorems remain inherited, unverified dependencies of this report. Applicability L4 denotes a strong methodological match, not a realized global proof or an algorithmic speedup. Contribution assessment: C2/5, limited source-specific local reformulations and proved extensions, with no new global Penrose theorem claimed.

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# 1 Scope, provenance and scientific status

The object is the October 5, 2026 composite in the locally pinned public OpenAI mathematics tree, not a generic Penrose conjecture. The catalogue identifies M512, family 260, applicability L4, and the pinned revision

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

The working source directory is recorded in the read ledger rather than silently replaced by a newer web version. The parent's `batch25-source-identity.json` receipt verifies 49 current local text files against pinned Git blobs after reading, and 50 unchanged blob pairs between the pinned and earlier October 8 cached trees. The cached target is `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`. This is not a fresh live fetch or a guarantee of unchanged bytes throughout reading. The audit freeze records current SHA256 values.

The 47 TeX files include scientific modules, wrappers, preamble and three figures. The 18-line README and 981-line bibliography were also read. Literal coverage is broad; proof verification is narrower: the propositions explicitly proved below and the bounded checks described in Section 13. No proof assistant, exhaustive dependency verification or global PDE solver was used. Statements about what the source proves describe the manuscript's argument, not independent certification of its conclusion.

The core, foundations and applications maps were used for routing only. Actual private mathematical inputs were read: Astra TR8, lines 1–320 (PDE-first domains, traces and inverses); Astra TR1, `stability.tex`, the full file (full variations and local inverse conditions); revised foundation TR250, lines 136–340 and 445–509 (moment coordinates, compatible complexes, physical metrics and a partially read hypercircle discussion); and final TR214, lines 321–510 (full nonlinear quasilinear variations and local Newton proofs). No theorem is credited to the unfinished hypercircle reading. The sidecar specifies exact paths and literal intervals. A routing summary does not substitute for these proof texts. No other report's contribution rating is inherited.

Private sources were kept local. External queries used only public paper titles, authors and identifiers. The primary-literature check on October 8, 2026 confirmed public records for Bray–Lee, Bray–Khuri, Khuri–Weinstein–Yamada, Huang–Lee, Lesourd–Unger–Yau, Khuri–Kopiński and Allen–Bryden–Kazaras–Khuri; it was not a full rereading of their proofs. The suboptimal-constant result [8] does not verify the sharp constant in this composite. A PDE reduction [3] is not an existence theorem for every similar coupled system.

# 2 Distinct geometric formulations

## 2.1 Neutral spatial dimension three

The source's conventions are

$$16\pi\mu = R_g + (\mathrm{tr}_g K)^2 - |K|_g^2, \qquad 8\pi J = \mathrm{div}_g(K - (\mathrm{tr}_g K)g).$$

Here $J$ is a covector and the dominant energy condition (DEC) is $\mu \geq |J|_g$. The basic exterior is connected and orientable, complete up to a nonempty smooth compact boundary, with one asymptotically flat end. The inner normal $\nu$ points into the end, and $\theta_+ = H + \mathrm{tr}_\Sigma K \leq 0$. Decay $g - \delta = O_2(r^{-q})$, $K = O_1(r^{-1-q})$, $q > 1/2$, and integrable constraints belong to the formulation.

The mass is $m = \sqrt{E^2 - |P|^2}$, not $E$ before rest reduction. The source concludes $m > 0$ and $m \geq \sqrt{a/(16\pi)}$. The enclosing area $a$ uses full boundaries of admissible filled competitors. Boundary-coincident portions count; cuts may be disconnected and need not be trapped. An arbitrary selected MOTS area cannot replace this infimum.

The neutral bridge also permits a global trapped-side domain $D_+$ with multiple components or ends. A competitor $D'_+ \subset D_+$ containing the selected tail is not automatically required to omit all other tails. Removing other tails gives a one-sided inclusion of classes and a one-sided comparison of infima, not equality of classes.

### 2.2 Spatial dimension four

The normalization changes:

$$\mu = \tfrac{1}{2}(R_g + (\operatorname{tr} K)^2 - |K|^2), \quad J = \operatorname{div}(K - (\operatorname{tr} K)g), \quad \omega_3 = 2\pi^2.$$

The ADM energy and momentum denominators are $6\omega_3$ and $3\omega_3$. The selected end has $q > 1$; finitely many other ends are allowed. The outer-domain formulation includes the selected tail and excludes all other tails, counting the full boundary and coincident parts. This differs from the neutral three-dimensional bridge.

The numerical conclusion is

$$m \geq \tfrac{1}{2}(A/\omega_3)^{2/3}.$$

The Tangherlini model $g = (1 + m/(2r^2))^2\delta$ has $A = \omega_3(2m)^{3/2}$. For positivity of the infimum the source uses a closed three-form in a tube, of bounded comass and integral one on every separator. The numerical Riemannian input is Bray–Lee [2]; a spin hypothesis from a different equality argument should not be added to this numerical theorem.

### 2.3 Charge and two area branches

In spatial dimension three define

$$\mu_{\mathrm{m}} = \mu - \frac{|E|^2 + |B|^2}{8\pi}, \qquad J_{\mathrm{m}} = J - \frac{(E \times B)^\flat}{4\pi}.$$

The assumptions include the matter DEC $\mu_{\mathrm{m}} \geq |J_{\mathrm{m}}|$, source-free fields, stated falloff, and finite charges. For $Q = (Q_E^2 + Q_B^2)^{1/2}$ and $r = \sqrt{a/(4\pi)}$, the source retains both conclusions

$$m \geq Q, \qquad r \leq m + \sqrt{m^2 - Q^2}. \tag{1}$$

For $r \geq Q$, this is equivalent to $m \geq (r + Q^2/r)/2$. For $r < Q$, the radius bound is automatic once $m \geq Q$, and that stronger mass-function bound does not follow.

Indeed $r^2 - 2mr + Q^2 \leq 0$ means $r_- \leq r \leq r_+$, where $r_\pm = m \pm \sqrt{m^2 - Q^2}$. Since $r_- \leq Q \leq r_+$, the lower root condition is automatic on $r \geq Q$. For $Q = m = 1$, $r = 1/2$, (1) holds but the mass-function right side is $5/4$. This exact counterexample forbids the branch extension.

The electric finite-end variant assumes $B = 0$, zero selected-end momentum, and a separating full-boundary quantity $a_0$ with $r_0 = \sqrt{a_0/(4\pi)} \geq |Q_0|$. The balance is $Q_0 = Q(S) - \sum_{j \neq 0} Q_j$. The compact boundary $S$ may be empty if other ends supply separation. The charged Riemannian endpoint is imported; the composite also cites the separate original-data charged manuscript, whose proof is not read or certified here. The classical multiple-black-hole theorem [4] is context, not a substitute for checking these exact endpoint hypotheses.

# 3 Reduction and preparation

## 3.1 Strict DEC and end charges

In dimension three strictification uses $g_s = u^4 g,\ K_s = u^2 K,\ u = 1 + s\phi$, with a positive profile, $\partial_\nu \phi = -1$, and $-\Delta\phi - |K||\nabla\phi| \geq w > 0$. It increases energy by the end coefficient, keeps momentum fixed, and does not decrease enclosing area. Non-timelike original data are treated by limits of strict timelike data, not by boosting a null energy-momentum vector.

The actual far-annulus linearized constraints are

$$Lh = \partial_i \partial_j h_{ij} - \Delta \operatorname{tr} h, \qquad (Dk)_i = \partial_j (k_{ij} - (\operatorname{tr} k)\delta_{ij}).$$

Affine scalar moments and Killing-vector moments obstruct compact support inversion. The source uses compatible inverses, charge adjustment, vacuum bending and radial DEC repair. The remaining deficits scale as $o(R^{-3})$ in dimension three and $o(R^{-4})$ in dimension four. A small component residual without the appropriate annular scaling and obstruction moments is not that repair.

In dimension four a positive shell $g_u = u^2 g,\ K_u = uK$ raises energy by $2\lambda$ and preserves momentum. It does not give a uniform boost near null energy-momentum. The proper radial compression $\Phi_R^* g \leq (1 + o(1)) g_R$ controls every competing cut, rather than one presumed minimizer.

## 3.2 Two frontiers and a nonsymmetric operator

Prepared black and white faces have expansions $H + P = c_b$, $H - P = c_w$, $c_b, c_w < 0$; either color may be absent. Right-continuity of total trapped regions and the collar choices matter. The offset $C$, $h = \eta f$, and $\rho > 0$ are arranged so that

$$|(h + C) \operatorname{tr} K| + |\nabla C| + \rho$$

uses only a prescribed fraction of the strict DEC margin. The source transports Jacobi supports through an exact collar isometry; arbitrary transport with uncontrolled second derivatives does not give the same error estimate.

The face stability operator is genuinely nonself-adjoint:

$$L_\Sigma v = -\Delta_\Sigma v + 2K(\nu, \nabla_\Sigma v) + \{\operatorname{tr}_\Sigma(\nabla_\nu K) - |\mathrm{II}|^2 - \mathrm{Ric}(\nu, \nu)\} v. \tag{2}$$

At a zero principal eigenvalue a bordered solve uses positive right and left eigenfunctions, not an invented positive energy form.

**Proposition 3.1** (Bordered local inverse)**.** *On a compact connected smooth face suppose* (2) *has index zero, simple kernel* $\operatorname{span}\{\phi\}$*, adjoint kernel* $\operatorname{span}\{\phi^*\}$*, and* $\phi, \phi^* > 0$*. Let* $\ell(v) = \int_\Sigma v$*. Then*

$$\mathbb{B}(v, a) = (L_\Sigma v - a, \ell(v)) : C^{2,\alpha}(\Sigma) \times \mathbb{R} \longrightarrow C^{0,\alpha}(\Sigma) \times \mathbb{R}$$

*is a bounded isomorphism. If* $T$ *inverts* $L_\Sigma$ *from its range into* $\{\ell(v) = 0\}$*, then*

$$a = -\frac{\langle \phi^*, f \rangle}{\langle \phi^*, 1 \rangle}, \qquad v = T(f + a) + \frac{b}{\ell(\phi)} \phi$$

*solves* $\mathbb{B}(v, a) = (f, b)$*. If a bounded same-domain perturbation* $\mathbb{E}$ *satisfies* $\theta = \|\mathbb{B}^{-1}\mathbb{E}\| < 1$*, then* $\|(\mathbb{B} + \mathbb{E})^{-1}\| \leq \|\mathbb{B}^{-1}\|/(1 - \theta)$.

*Proof.* The Fredholm range annihilates $\phi^*$. Pairing $Lv - a = f$ forces the displayed $a$, with positive denominator. Removing the mean kills the kernel since $\ell(\phi) > 0$. The Fredholm alternative and bounded inverse theorem give $T$. The formula has mean $b$ and solves the equation. A homogeneous pair has $a = 0$, $v = c\phi$, and zero mean, hence vanishes. Finally factor $\mathbb{B} + \mathbb{E} = \mathbb{B}(I + \mathbb{B}^{-1}\mathbb{E})$ and use the norm-convergent Neumann series. □

This classical local specialization does not supply a numerical inverse bound for the source's actual faces, or a uniform bound over changing faces. A disconnected union needs one scalar and normalization per zero-mode component.

## 4 The actual graph–conformal system

Use $c = 4$ in spatial dimension three and $c = 3$ in dimension four; $d$ below is a scalar, not dimension. Fix prepared $g, K, C, \rho$, a smooth nonincreasing cutoff $\vartheta$, and

$$p(t) = n\vartheta(nt), \quad l(t) = \exp\left(\int_0^t p(s)\,\mathrm{d}s\right), \quad \ell = e^{-\epsilon n}, \quad \eta = \ell^{3/2}.$$

Here $\vartheta = 1$ to the left of zero and 0 to the right of one; $n \geq 4$ is sufficiently large. Define

$$s = \nabla f, \quad D = 1 + l^2|s|^2, \quad d = D^{-1}, \qquad w = ls/\sqrt{D}, \quad v = |w|^2 = 1 - d, \tag{3}$$

$$Z = t + \frac{\log D}{2c}, \qquad \chi = \frac{cd}{c + pv}, \quad b = \frac{c + p}{c + pv}, \tag{4}$$

$$A_d = I - w \otimes w, \qquad A_\chi = I - bw \otimes w, \tag{5}$$

$$u = e^{2t} l\sqrt{D}, \qquad H^f = \frac{l}{\sqrt{D}} \operatorname{Hess} f, \quad S = K + H^f. \tag{6}$$

For fixed $s$, $Z_t = 1 + pv/c > 0$, and the respective limits at the two ends of the $t$-axis are infinite. Thus $t = t(Z, s)$ is smooth for all finite inputs without a floor hypothesis. The direction-free expressions are smooth at $s = 0$. The axial eigenvalue $\chi$ is positive for finite inputs; it is not uniformly positive over all inputs.

The endpoint system on $\Omega_R$ is

$$F := A_\chi : S = \eta f + C =: F_{\mathrm{p}}, \tag{7}$$

$$V = uA_\chi(c\nabla Z + K(w, \cdot)), \quad \operatorname{div} V = u\Xi, \tag{8}$$

$$\Xi = \delta_0 \mathcal{T} + \rho + \delta_0 \eta\, l|s|^2/\sqrt{D} - m_0(t) C_n(\rho_0 + v\rho_1), \tag{9}$$

$$m_0(t) = \vartheta(n(t + \epsilon)). \tag{10}$$

The small $\delta_0$, polynomial coefficient $C_n$, and tails are prepared quantities. In dimension three $\rho_0$ has $r^{-4}$ scale and $\rho_1 = r^{-2\gamma}$, $3/4 < \gamma < 1$; in dimension four the scales are $r^{-4-\delta_1}$ and $r^{-3}$. These losses are not compactly supported.

On the outer sphere, $f = Z = 0$. On the inner faces, $f = b_\pm/\eta$ and

$$V \cdot \nu = q_B = uB_0, \quad B_0 = -H + w_\nu(F_{\mathrm{p}} - P_B) - N_B d, \quad P_B = \operatorname{tr}_B K, \tag{11}$$

where $N_B > 1 + \sup|H|$. The flux uses the prescribed $F_{\mathrm{p}}$, not a Hessian trace recomputed from unconstrained trial states. Its Newton boundary row has no independent $\operatorname{Hess} f$ term.

Set $\bar g = g + l^2\,\mathrm{d}f \otimes \mathrm{d}f$. Then $\hat g = e^{4t}\bar g$ in dimension three and $\hat g = e^{2t}\bar g$ in dimension four. The scalar identity is

$$\tfrac{1}{2}e^{\kappa t}R_{\hat g} = \mathfrak{m} + \mathcal{T} + w(F) - F\,\mathrm{tr}\,K - u^{-1}\,\mathrm{div}\,V, \tag{12}$$

with $(\kappa, \mathfrak{m}) = (4, 8\pi(\mu + J(w)))$ or $(2, \mu + J(w))$, respectively.

The exact smooth quadratic is conveniently written

$$k = -H^f - p(\,\mathrm{d}t \otimes w^\flat + w^\flat \otimes \mathrm{d}t), \quad Q = K - k, \tag{13}$$

$$\mathcal{B}(A,B) = \langle A, B\rangle - (\mathrm{tr}\,A)(\mathrm{tr}\,B), \quad P_Q = Q - (\mathrm{tr}\,Q)g, \tag{14}$$

$$\mathcal{T} = \tfrac{1}{2}\mathcal{B}(Q,Q) + 2P_Q(w, \nabla t) + c(p+1)|\,\mathrm{d}t|^2_{A_d} + FG, \tag{15}$$

$$G = F + bS(w,w) + 2(p+1)w(t).$$

Here $Q$ is a tensor, not total charge. For nonzero slope take $e = s/|s|$, $j = S(e,e)$, $M = S(e,\cdot)|_{e^\perp}$, $T = S|_{e^\perp}$, $x = e(t)$, $y = \nabla_\perp t$, $a = |w|$. The three-dimensional square completion is

$$\begin{aligned}\mathcal{T} = {}& \tfrac{1}{2}|T^{\mathrm{tf}}|^2 + |M + (p+1)ay|^2 + [4(p+1) - v]|y|^2 \\ & + 4(p+1)d\left(x + \frac{\chi a j}{4d}\right)^2 + \tfrac{3}{4}\left(F - \frac{\chi j}{3}\right)^2 + \frac{4d(9-d)}{3(4+pv)^2}j^2.\end{aligned} \tag{16}$$

In spatial dimension four it is

$$\begin{aligned}\mathcal{T} = {}& \tfrac{1}{2}|T^{\mathrm{tf}}|^2 + |M + (p+1)ay|^2 + [3(p+1) - v]|y|^2 \\ & + 3(p+1)d\left(x + \frac{\chi a j}{3d}\right)^2 + \tfrac{2}{3}\left(F - \frac{\chi j}{4}\right)^2 + \frac{d(48-3d)}{8(3+pv)^2}j^2.\end{aligned} \tag{17}$$

At zero slope, $\mathcal{T} = \frac{1}{2}(|S|^2 + (\mathrm{tr}\,S)^2) + c(p+1)|\,\mathrm{d}t|^2$. These are pointwise geometric squares, not strong monotonicity or invertibility of the entire coupled PDE map.

# 5 Full nonlinear variation and Ram realization

## 5.1 Exact joint derivative

Fix the background, faces and preparation parameters. Vary the actual unknowns by $(\varphi, \zeta)$, put $h = \nabla\varphi$, $\alpha = \delta t$. Implicit differentiation gives

$$\alpha = \frac{c\zeta - l^2\langle s, h\rangle/D}{c + pv}, \tag{18}$$

$$\delta D = 2l^2(\langle s, h\rangle + p|s|^2\alpha), \qquad \delta d = -d^2\delta D, \quad \delta p = p'(t)\alpha, \tag{19}$$

$$\delta w = \frac{l}{\sqrt{D}}h + \left(p\alpha - \frac{\delta D}{2D}\right)w, \qquad \delta v = 2\langle w, \delta w\rangle, \tag{20}$$

$$\delta b = \frac{(c+pv)\delta p - (c+p)(v\delta p + p\delta v)}{(c+pv)^2}, \qquad \delta A_\chi = -\delta b\, w\otimes w - b(\delta w\otimes w + w\otimes\delta w), \tag{21}$$

$$\frac{\delta u}{u} = (p+2)\alpha + \frac{\delta D}{2D}, \qquad \delta H^f = \frac{l}{\sqrt{D}}\,\mathrm{Hess}\,\varphi + \left(p\alpha - \frac{\delta D}{2D}\right)H^f. \tag{22}$$

With $\delta S = \delta H^f$, the full trace and flux derivatives are

$$\delta F = \delta A_\chi : S + A_\chi : \delta S, \tag{23}$$

$$\begin{aligned}\delta V = {}& (\delta u\, A_\chi + u\,\delta A_\chi)(c\nabla Z + K(w,\cdot)) \\ & + uA_\chi(c\nabla\zeta + K(\delta w,\cdot)).\end{aligned} \tag{24}$$

These formulas do not differentiate $|s|$ at zero; their smooth direction-free form handles that point. For completeness write $z = \nabla t$, $z' = \nabla\alpha$, and let primes on the following tensors denote variations:

$$\begin{aligned} Q' &= \delta S + \delta p(z^\flat \otimes w^\flat + w^\flat \otimes z^\flat) + p(z'^\flat \otimes w^\flat + w^\flat \otimes z'^\flat) \\ &\quad + p(z^\flat \otimes \delta w^\flat + \delta w^\flat \otimes z^\flat), \qquad (25) \\ A'_d &= -\delta w \otimes w - w \otimes \delta w, \qquad (26) \\ G' &= \delta F + \delta b\, S(w,w) + b\{\delta S(w,w) + 2S(\delta w, w)\} \\ &\quad + 2\delta p\, \langle w, z\rangle + 2(p+1)(\langle \delta w, z\rangle + \langle w, z'\rangle), \qquad (27) \\ \delta\mathcal{T} &= \mathcal{B}(Q, Q') + 2P_{Q'}(w,z) + 2P_Q(\delta w, z) + 2P_Q(w, z') \\ &\quad + c\delta p\, \langle A_d z, z\rangle + c(p+1)(\langle A'_d z, z\rangle + 2\langle A_d z, z'\rangle) + \delta F\, G + F G'. \qquad (28) \end{aligned}$$

The spatial derivative $z' = \nabla\alpha$ includes every coefficient in (18); freezing them omits nonlinear terms. For $W = l|s|^2/\sqrt{D}$,

$$\delta W = \frac{l}{\sqrt{D}}\left[2\langle s, h\rangle + \left(p\alpha - \frac{\delta D}{2D}\right)|s|^2\right], \qquad (29)$$

$$\delta\Xi = \delta_0\delta\mathcal{T} + \delta_0\eta\delta W - C_n m_0'(t)\alpha(\rho_0 + v\rho_1) - C_n m_0(t)\rho_1\delta v. \qquad (30)$$

On the inner boundary the correct variation is

$$\delta B_0 = \delta w_\nu(F_{\mathrm{p}} - P_B) + w_\nu\eta\varphi - N_B\delta d, \qquad \delta q_B = \delta u\, B_0 + u\,\delta B_0. \qquad (31)$$

Homogeneous variations have $\varphi = 0$ at both boundaries, $\zeta = 0$ only at the outer boundary.

**Proposition 5.1** (Complete endpoint differential)**.** *For sufficiently smooth states and variations on a fixed smooth truncation,* (18)–(31) *give the full Fréchet derivative of* (7)–(11)*. They have the usual product/composition bounds on bounded smooth sets. They are not a frozen-coefficient derivative.*

*Proof.* Differentiate $2c(Z - t) = \log D$ and $D = 1 + l(t)^2|s|^2$, using $l' = pl$, and solve for $\alpha$. The denominator $c + pv$ is at least $c$. Inverse, square-root and product rules give the displayed coefficient derivatives. Differentiating $A_\chi : S$ and all factors of $V$ gives (23)–(24). Since $Q = S + p(z^\flat \otimes w^\flat + w^\flat \otimes z^\flat)$, its derivative is $Q'$; bilinearity in (15) then yields every term of (28). The cutoff loss varies both through $m_0(t)$ and through $v$, giving both final terms of (30). Finally the boundary trace is prescribed as $F_{\mathrm{p}} = \eta f + C$; its variation is $\eta\varphi$, proving (31). Smooth composition proves the local regularity assertion. □

## 5.2 Weak domain and the inner normal

Let $\mathcal{V} = \{\psi \in H^1(\Omega_R) : \psi|_{S_R} = 0\}$. At suitably regular states set

$$\mathcal{R}_2(f, Z)[\psi] = \int_{\Omega_R} V \cdot \nabla\psi + \int_{\Omega_R} u\Xi\psi + \int_B q_B\psi. \qquad (32)$$

The outward normal is $-\nu$ on $B$. Integration by parts therefore proves equivalence with $\operatorname{div} V = u\Xi$, $V_\nu = q_B$ for smooth solutions. The last sign is positive. The differential is

$$D\mathcal{R}_2[\varphi, \zeta; \psi] = \int \delta V \cdot \nabla\psi + \int (\delta u\,\Xi + u\,\delta\Xi)\psi + \int_B \delta q_B\psi.$$

The first residual is $F - \eta f - C$, with derivative $\delta F - \eta\varphi$. This is not asserted to define a nonlinear map on unrestricted $H^1 \times H^1$; its Hessian terms need stronger state regularity. Smooth or suitable Hölder states with weak tests suffice for the identities. A stronger Sobolev formulation needs its own mapping proof.

A legitimate coercive subproblem fixes uniformly positive $A_\chi$ and prescribed $B_1, Q_1, q_1$. Then $\operatorname{div}(cA_\chi \nabla Y + B_1) = Q_1$, inner $\nu$-flux $q_1$, and zero outer trace give

$$c \int A_\chi \nabla Y \cdot \nabla \psi = - \int Q_1 \psi - \int B_1 \cdot \nabla \psi - \int_B q_1 \psi.$$

Poincaré and Lax–Milgram provide its mixed-boundary inverse on $\mathcal{V}'$, as in the actual Astra TR8 proof. It is not the inverse of the entire nonlinear system. All-Neumann data require compatibility and a gauge per component. A free-space kernel does not impose these boundary realizations.

### 5.3 Physical finite moments, not a surrogate PDE

For a conforming finite trial family $u_h = \sum_j a_j e_j$, take unisolvent bounded moments $m_i = \ell_i(u_h)$, $E_{ij} = \ell_i(e_j)$, $a = E^{-1}m$. For a fixed invertible residual coordinate map $S_0$,

$$\mathcal{R}_{\mathrm{Ram}}(m) = S_0 \mathcal{R}_h(E^{-1}m), \qquad D\mathcal{R}_{\mathrm{Ram}} = S_0 D\mathcal{R}_h(E^{-1}m)E^{-1}. \tag{33}$$

The residual uses the actual equations above, their lifts and full boundary terms, with exact integration when exact equivalence is claimed. The trial basis must have the required differentiability; an arbitrary $H^1$ basis cannot be inserted into Hess $f$.

**Proposition 5.2** (Scope of finite Ram equivalence)**.** *For fixed invertible $E, S_0$,* (33) *preserves finite roots and Jacobian rank. If $G$ is the physical trial Gram, its moment form is $E^{-T}GE^{-1}$. For compatible differentials $D_{k+1}D_k = 0$ and invertible synthesis $R_k$, the transformed maps $D_k^{\mathrm{R}} = R_{k+1}^{-1} D_k R_k$ satisfy $D_{k+1}^{\mathrm{R}} D_k^{\mathrm{R}} = 0$, with transported physical adjoints. None of these statements supplies a continuum inverse.*

*Proof.* Invertible multiplication preserves zeros and rank. Substitution in $a^T G a$ gives the Gram expression. The adjacent $R_{k+1}R_{k+1}^{-1}$ cancels in the product of differentials. Adjoints follow from these preserved inner products, not from an unweighted transpose. These finite identities imply neither a uniform inverse bound on a refining family nor completeness. □

This is the actual compatible-coordinate argument in revised TR250. Curved flux elements retain $J/\det J$; one-form pullbacks retain $J^T$, with physical metric and volume factors. State-dependent moment banks introduce derivatives of the banks. TR214's actual quasilinear example gives the same warning: the variation of $u^2 + \mu_2[(u')^2 + uu'']$ is

$$2u\varphi + \mu_2[2u'\varphi' + u''\varphi + u\varphi''].$$

Differentiated rows also differentiate $\mu_2$.

For clarity, Astra TR1's local inverse result is conditional. If $\mathcal{R} : X \to Y$ is $C^1$ on $B(x_0, r)$, $B : Y \to X$ is a bounded isomorphism, and

$$\eta_0 = \|B\mathcal{R}(x_0)\|, \quad q_0 = \sup_{B(x_0,r)} \|I - BD\mathcal{R}\| < 1, \quad \eta_0 + q_0 r \le r,$$

then the mean-value formula shows $x \mapsto x - B\mathcal{R}(x)$ is a contraction of that ball. Its unique fixed point is a zero, and its distance from $x_0$ is at most $\eta_0/(1-q_0)$. No such continuum $B, \eta_0, q_0, r$ is established here. A rectangular left inverse of sampled rows can annihilate a nonzero residual, so a finite matrix check is not that theorem.

## 6 Analytic dependencies and mass limits

The source's floor avoidance first obtains coarse estimates and only then the narrow height estimate excluding $t = -\epsilon$. Using the narrow estimate to prove itself would be circular. The scalar trace solve $f[Z]$ must exist over the whole degree trial space, not just at fixed points. Its large-slope axial degeneracy has powers $|s|^{-8/(n+4)}$ in dimension three and $|s|^{-6/(n+3)}$ in dimension four, in the stated fixed-parameter regime. Logarithmic barriers and boundary estimates are essential.

The three-dimensional argument uses axial Cordes estimates, Bochner identities and Morrey control. The four-dimensional proof has a separate gradient-drop/affine-alternative argument; copying a three-dimensional embedding exponent is not sufficient. For a fixed axis $e$, the distributional observation

$$\Delta_\perp Y + \chi(x)\partial_{ee} Y = 0 \implies \Delta_\perp(\partial_e Y) + \partial_e\{\chi\, \partial_e(\partial_e Y)\} = 0$$

does not require separately differentiating measurable $\chi$ as a classical function. A varying axis introduces additional terms, so that extension cannot be assumed.

The other scalar equation has natural quadratic gradient growth and a Morrey-controlled measure term. Even inner reflection creates the surface jump $2Q_{\mathrm{normal}}$; zero-Dirichlet outer reflection is odd. The two exponential changes, Green estimates, Hölder bounds and alternating Schauder arguments remain inherited analytic steps. Leray–Schauder degree one proves existence only after the bounds and compactness are established; it does not prove uniqueness of the coupled solution.

From the exact inner identity

$$V_\nu/u = d(H + c\partial_\nu t) - H + w_\nu(F - P_B),$$

the boundary condition forces $H + c\partial_\nu t = -N_B$, so the conformal graph boundary has the intended negative mean curvature. On the exterior solution,

$$\mathcal{F}_\infty = \int_\Omega u\Xi + \int_B V \cdot \nu. \tag{34}$$

The inner sign is positive. The energy change is $-\mathcal{F}_\infty/(8\pi)$ in dimension three and $-\mathcal{F}_\infty/(3\omega_3)$ in dimension four. The latter uses $t, Z = O_2(r^{-2})$, $V = 3\nabla t + O(r^{-5})$; the former has $V = 4\nabla t + O(r^{-3})$.

The scalar loss is bounded by $\mathrm{poly}(n)(\ell + \ell^2/\eta) = \mathrm{poly}(n)(\ell + \ell^{1/2})$. The order is outer radius to infinity at fixed $n$, then $n \to \infty$ at fixed $\epsilon$, then $\epsilon \downarrow 0$; strictification and rest reduction have additional fixed-scale limits. An arbitrary numerical diagonal sequence does not replace the uniform estimates. Unwanted-end truncations in dimension four also require control of their enclosing infima.

If $t \geq -\epsilon$, the metric lower bound loses at most $e^{-4\epsilon}$ in two-dimensional area and $e^{-3\epsilon}$ in three-dimensional volume. The latter becomes $e^{-2\epsilon}$ under the 2/3 mass power. The following elementary statement applies to every cut.

**Proposition 6.1** (Full-infimum metric sensitivity)**.** *For a fixed admissible class $\mathcal{C}$ of rectifiable $k$-dimensional cuts, including all coincident portions, if $0 < \lambda g \leq \widetilde{g} \leq \Lambda g$, then*

$$\lambda^{k/2} a_g \leq a_{\widetilde{g}} \leq \Lambda^{k/2} a_g, \qquad a_g = \inf_{\Sigma \in \mathcal{C}} |\Sigma|_g.$$

*The same holds under a bijective diffeomorphic correspondence of classes, using the pullback metric.*

*Proof.* On each approximate tangent $k$-plane, relative metric eigenvalues lie in $[\lambda, \Lambda]$. The area density is the square root of their product. Integrate these bounds on every cut and take infima. For the upper bound use any minimizing sequence; existence of a minimizer is unnecessary. Pullback preserves each area integral exactly. □

This compares metrics on the same class, not different end or topological admissibility conditions.

# 7 Charge comparison and period loss

For a divergence-free field $X$, its flux form is $\alpha_X = \iota_X \, \mathrm{d}V_g$, and $\mathrm{d}\alpha_X = 0$. Keeping it fixed under $g' = u^4 g$ sends $X$ to $u^{-6}X$. Keeping vector components fixed would generally lose flux preservation. For $|Y| \leq 1$, define

$$I(Y) = |E|^2 + |B|^2 + 2\langle E \times B, Y \rangle, \qquad \mathcal{D}(Y) = 8\pi(\mu + J(Y)) - I(Y).$$

Its infimum is $8\pi(\mu_{\mathrm{m}} - |J_{\mathrm{m}}|)$. The conformal charged identity is

$$u^4 \mathcal{D}_{g'}(u^{-2}Y) = \mathcal{D}_g(Y) + (1 - u^{-4})I(Y) + 4u^{-1}\{-\Delta u + K(\nabla u, Y)\}.$$

The electromagnetic $r^{-4}$ tail is therefore not a compact vacuum deficit.

For total $Q > 0$, rotate by

$$O = \frac{1}{Q} \begin{pmatrix} Q_E & Q_B \\ -Q_B & Q_E \end{pmatrix}$$

to fields $X_e, X_o$ with total charges $Q, 0$. The square

$$I(w) = |X_o + w \times X_e|^2 + |X_e|^2 - |w \times X_e|^2 \geq |\alpha_e|^2_{\hat{g}}$$

permits retaining forms $(\alpha_e, 0)$ and rotating back. For $Q = 0$ this displayed rotation is undefined; the zero-charge comparison needs its own fixed rotation or zero retained field.

**Proposition 7.1** (Period preservation criterion)**.** *For $Q > 0$ and any oriented closed two-cycle $\Gamma$,*

$$\begin{pmatrix} \int_\Gamma \alpha_E \\ \int_\Gamma \alpha_B \end{pmatrix} - \begin{pmatrix} \int_\Gamma \alpha_E^{\mathrm{comp}} \\ \int_\Gamma \alpha_B^{\mathrm{comp}} \end{pmatrix} = O^T \begin{pmatrix} 0 \\ \int_\Gamma \alpha_o \end{pmatrix}.$$

*Both total charges are preserved. All periods on a specified family of cycles are preserved exactly when the discarded $\alpha_o$ has zero periods on that family. For the full real de Rham pairing this means $[\alpha_o] = 0$. Zero total charge alone is insufficient.*

*Proof.* Integration commutes with the constant inverse rotation. The discarded period at the distinguished end is zero by its definition. Since $O^T$ is invertible, its period difference vanishes on a cycle exactly when the discarded period does. The all-cycle statement is the de Rham pairing criterion. Opposite nonzero fluxes on separate components can sum to zero at the distinguished end, showing why the total charge condition alone does not give the stronger assertion. □

Retaining the discarded form would require an additional scalar-curvature budget for its norm; the one-field square does not provide one. The final comparison rescaling $\widetilde{g} = e^{4\epsilon}\hat{g} \geq g$ retains flux forms. Coordinates $z = e^{2\epsilon}y$ give $E_{\widetilde{g}} = e^{2\epsilon}E_{\hat{g}}$, while charge periods do not scale. This lossy comparison is not an original-data charged rigidity proof.

# 8 Original-data equality, not comparison equality

## 8.1 Full obstacle area and tied minimizers

The three-dimensional equality module starts with an outer-area-minimizing MOTS boundary, possibly with multiple components, and an outermost condition excluding relevant partial coincidences. The four-dimensional rigidity module assumes connected boundary and uses that fact to localize a touching tied minimizer. Full obstacle perimeter still counts coincident boundary portions.

The directional area derivative has the form

$$a'(0+) = \min_{\Sigma \in \mathcal{T}} \frac{1}{2} \int_{\Sigma} \operatorname{tr}_{\Sigma} h$$

over tied minimizing cuts. The next proposition isolates its elementary envelope part from the geometric compactness needed to apply it.

**Proposition 8.1** (Compact tied-minimizer envelope)**.** *Suppose a compact parameter space $\mathcal{K}$ contains the relevant competitors, each $A_t$ and $A_1$ is continuous, and $A_t = A_0 + tA_1 + o(t)$ uniformly on $\mathcal{K}$ as $t \downarrow 0$. If $a(t) = \min_{\mathcal{K}} A_t$, then*

$$a'(0+) = \min_{\{A_0 = a(0)\}} A_1.$$

*It suffices that all near-minimizers admit this compact parameterization and uniform expansion.*

*Proof.* Test at a minimizing competitor having least $A_1$ to obtain the upper derivative bound. Along any $t_j \downarrow 0$, select minimizers $\Sigma_j$ and pass to a convergent subsequence. Uniform convergence implies $A_0(\Sigma_*) = a(0)$. Since $A_0(\Sigma_j) - a(0) \geq 0$,

$$\frac{a(t_j) - a(0)}{t_j} \geq A_1(\Sigma_j) + o(1) \longrightarrow A_1(\Sigma_*) \geq \min_{\{A_0 = a(0)\}} A_1.$$

This proves the matching lower bound. □

The source must supply the compactness and first-variation continuity. Strict BV convergence alone does not show common tangent planes at intersecting tied boundaries. The perimeter union/intersection argument, regularity and Allard-type control are substantive prerequisites.

## 8.2 Active DEC, multipliers and KIDs

The supporting energy coefficients at equality are $b^0 = E/m,\ b = -P/m$, from the reverse Lorentz inequality. The active DEC constraints are $16\pi(\mu + J(v))$ or $2(\mu + J(v))$. Their unit ball varies with the metric: $\dot{v} = -\frac{1}{2} h^{\sharp} v$. Omitting this changes the adjoint.

The source separates infinite families of DEC and area constraints, obtaining positive multipliers on active directions, a probability measure on tied minimizers, and boundary/end multipliers. Regularized dust stresses, Bianchi identities, and common-plane localization remove unwanted multipliers. Boundary atoms are excluded with metric variations vanishing to first normal order but having free second normal jets. A finite cone or a first-jet test alone is not this argument.

The resulting smooth causal lapse and shift satisfy a KID system including

$$\operatorname{sym} \nabla X = -uK, \qquad \Delta u = -\operatorname{div} K(X),$$

and the Hessian equation involving $u(\mathrm{Ric} + (\operatorname{tr} K)K - 2K^2) - \mathcal{L}_X K$ and the dimension-normalized $X_{(i}J_{j)}$ term. They obey $u \geq |X|$, $u\mu + J(X) = 0$, and $u > 0$ in the interior. On the horizon,

$$X = u\nu, \qquad \partial_\nu u + K(X, \nu) = \kappa,$$

with $\kappa = 1/(4m)$ in dimension three and $\kappa = 1/\sqrt{2m}$ in dimension four. The stationary metric $-u^2\,\mathrm{d}t^2 + g(\,\mathrm{d}x + X\,\mathrm{d}t)^2$ induces the original $g, K$. A comparison metric with equal mass does not establish that. The multiplier regularity argument is inherited, not proved here; the public mass-minimizer work [5] is relevant context.

### 8.3 Null sets, static base and original embedding

Let $N = u^2 - |X|^2$. On $N > 0$,

$$h = g + N^{-1}X^\flat \otimes X^\flat, \quad h^{-1} = g^{-1} - u^{-2}X \otimes X, \quad \lambda = \sqrt{N}, \quad \mathcal{A} = N^{-1}X^\flat.$$

The source obtains null stress and vacuum on the positive region, then a twist equation for $\mathrm{d}\mathcal{A}$. It does not assume a smooth null-set boundary. Logarithmic weighted cutoffs, subtraction of an asymptotic exact linear form and end decay justify integration of twist. After twist vanishing, bounded $|\mathrm{d}\log N|_h$ excludes finite-distance internal null ends. Positivity and connectedness of the relevant component also require maximum-principle arguments.

Where $u > 0$ on the horizon, $\lambda = \sqrt{N}$ is the good base coordinate with even reflection. Where $u = 0$, $u = sa$, $X = s^2Y$, and $N = s^2(a^2 - s^2|Y|^2)$. Dividing a geometric normal by $u$ there is not legitimate. The static equations are

$$\lambda\,\mathrm{Ric}_h = \mathrm{Hess}_h\,\lambda, \quad \Delta_h\lambda = 0, \quad R_h = 0.$$

The auxiliary metric is $\lambda^2 h$ in dimension three and $\lambda h$ in dimension four. Their harmonic expansions, inversion regularity and doubling arguments differ. The four-dimensional $C^2$ compactification needs monopole, dipole and quadrupole information; a generic $W^{2,2}$ bound does not imply $C^0$ regularity in dimension four.

The arbitrary-ends positive mass input [6] has a distinguished asymptotically Schwarzschild end and dimension restrictions. The source's additional $C^2$-rigidity step uses smoothing, local Ricci variation, positive integrated scalar curvature and $-6\Delta f + Rf = 0$. Auxiliary truncated ends have Neumann conditions, with the distinguished end anchored. Its precise flux is what yields the mass contradiction.

Euclidean doubling then identifies a round boundary and a Schwarzschild or Tangherlini static base. Simple connectedness makes the closed time connection globally exact. Where $u > 0$, $T_0 = -(2\kappa)^{-1}\log N + B$ gives Kruskal combinations $U = -ND_*(N)e^{-\kappa B}$, $V = D_*(N)e^{\kappa B}$. Where $u = 0$, finite $T_0$ and signed smooth $\lambda$ replace the logarithmic chart. Smoothness in original angular variables relies on the previously proved even reflection. The embedding recovers the original exterior, not arbitrary data behind the horizon. The dimension-four converse derives an asymptotically affine spacelike end before computing boosted charges. None of these global equality steps follows merely from equality of a limiting comparison inequality.

# 9 Local Schwarzschild–AdS analysis

## 9.1 A fixed seed and an actual branch

Now spatial dimension is three and $\Lambda = -3$. Let $m_0 > 0$, $1 + a^2 - 2m_0/a = 0$, and

$$g_0 = \mathrm{d}s^2 + r(s)^2\sigma, \quad r(0) = a, \quad r' = N > 0, \quad N^2 = 1 + r^2 - \frac{2m_0}{r}.$$

A fixed smooth transverse-traceless tensor $q$ lies in $C^{1,\alpha}_\tau$, $3/2 < \tau < 3$. The module assumes a given small positive branch near one:

$$\Delta\phi = \tfrac{3}{4}(\phi^5 - \phi) - \tfrac{\varepsilon^2}{8}|q|^2\phi^{-7}, \qquad \partial_s\phi|_S = \tfrac{\varepsilon}{4}q_{ss}\phi^{-3}, \tag{35}$$

$$g_\varepsilon = \phi^4 g_0, \qquad K_\varepsilon = \varepsilon\phi^{-2}q. \tag{36}$$

It is not a uniform branch-existence theorem over all seeds. The data are maximal vacuum, $R + 6 = |K|^2$, with MOTS boundary and $H = \varepsilon\phi^{-6}q_{ss}$. The main local result does not impose its sign. The outermost, outer-area-minimizing corollary has additional hypotheses.

For the AH mass covector $p_\mu$, let $m_{\mathrm{AH}} = (p_0^2 - \sum_i p_i^2)^{1/2}$ on its future timelike branch. Its spatial entries are not angular momenta. With the actual boundary area $A$, the deficit is

$$\mathcal{D}_q(\varepsilon) = m_{\mathrm{AH}} - b_*(A), \qquad b_*(A) = \sqrt{A/(16\pi)}(1 + A/(4\pi)).$$

The primary perturbative work [7] is genuine context, but its positive-mean-curvature and particular perturbation hypotheses should not be identified with every seed in the composite.

## 9.2 Jets and the mass formula

Put $L = \Delta_{g_0} - 3$, $\psi = \phi - 1$. Exactly,

$$L\psi = \tfrac{15}{2}\psi^2 + \tfrac{15}{2}\psi^3 + \tfrac{15}{4}\psi^4 + \tfrac{3}{4}\psi^5 - \tfrac{\varepsilon^2}{8}|q|^2(1+\psi)^{-7}.$$

For $\psi = \varepsilon u_1 + \varepsilon^2 u_2 + \cdots$,

$$Lu_1 = 0, \quad \partial_s u_1 = q_{ss}/4, \qquad Lu_2 = \tfrac{15}{2}u_1^2 - |q|^2/8, \quad \partial_s u_2 = -3q_{ss}u_1/4.$$

Finite jets and remainder bounds come from comparison, not an unproved analytic-dependence assumption. For $3/2 < \beta < \min(\tau, \sqrt{3})$, $e^{-\beta s}$ has strict bulk margin $3 - \beta^2$, with sufficiently small Robin coefficient. Finite-angular radial equations have roots $1, -3$; the decaying root is used for the refined nullspace jets.

For $W_0 = N$, $LW_0 = 0$, and $W_i = rx_i$, $LW_i = -3m_0 r^{-2}x_i$, the mass is identified through

$$p_\mu - p_\mu^{(0)} = \frac{1}{2\pi}\left[\int_S (\psi\,\partial_s W_\mu - W_\mu\partial_s\psi) - \int_M (W_\mu L\psi - \psi L W_\mu)\right]. \tag{37}$$

Integrability has radial scale $e^{(3-2\beta)s}$. One first identifies the actual fixed-parameter flux, then expands the function/source/boundary triple in integrable norms. Differentiating an unspecified remainder at infinity is not valid.

Rotational polynomial averaging gives finite-angular TT approximants, preserving a boundary sign when present and converging in the weaker $\beta$ norm. It need not produce nonlinear branches for those approximants. The quadratic coefficient includes

$$c_2 = a_0^{(2)} - \frac{\sum_i (a_i^{(1)})^2}{2m_0} - b_*'(A_0)A_2 - \tfrac{1}{2}b_*''(A_0)A_1^2.$$

Omitting spatial mass terms changes invariant mass into energy.

### 9.3 CMC jets and boost modes

Finite-order CMC surface jets are not automatically a global CMC foliation. The sphere Jacobi operator is

$$J_s = -\Delta_{r^2\sigma} - 2/r^2 + 6m_0/r^3.$$

For degree $j \geq 1$, its eigenvalue is $(j(j+1)-2)/r^2 + 6m_0/r^3$. For $j = 1$ this is positive but tends to zero at infinity. Thus there is no uniform unweighted gap. The rescaled boost block and higher modes are treated separately. The mass-aspect weight $k^{-3}$ transforms a Lorentz covector, so a zero-spatial-mass frame gives the Lorentz-norm Hawking limit. The quadratic $FJ_sF$ contribution is nonnegative after sphere integration and the stated mode restriction, not as an arbitrary pointwise expression.

## 10 A weighted TT identity and defect extension

Set

$$B = \mathrm{Ric}_{g_0} + 3g_0, \quad \mathrm{Hess}\, N = NB, \quad \mathrm{tr}\, B = 3, \quad \mathrm{div}\, B = 0, \quad \kappa = (1+3a^2)/(2a).$$

The radial and spherical eigenvalues of $B$ are $1 - 2m_0/r^3$ and $1 + m_0/r^3$. On $S$, define the positive operator

$$\mathcal{D} = -\Delta_{a^2\sigma} + 2\kappa/a.$$

Take the decaying solution $Lv = 0$ with $v|_S = \mathcal{D}^{-1}q_{ss}$, and $\mathcal{Q}(v) = \mathrm{Hess}\, v - vB$. Exterior existence is supplied by the source in its decay spaces; the next identities assume that solution and the indicated integrability and end behavior.

**Proposition 10.1** (TT lift and least weighted norm)**.** *Suppose $Lv = 0$ is smooth with adequate decay and $\mathcal{D}v = q_{ss}$ on $S$. Then $\mathrm{tr}\,\mathcal{Q}(v) = 0$, $\mathrm{div}\,\mathcal{Q}(v) = 0$, and $\mathcal{Q}(v)_{ss} = \mathcal{D}v$. For every adequately decaying smooth TT $k$ with $k_{ss} = q_{ss}$, and vanishing end term in the proof,*

$$\int_M N|k|^2 - \kappa \int_S v\mathcal{D}v = \int_M N|k - \mathcal{Q}(v)|^2. \tag{38}$$

*Thus $\mathcal{Q}(v)$ uniquely minimizes weighted norm over this TT boundary-trace fiber, and is orthogonal to its zero-trace TT difference space.*

*Proof.* The trace is $Lv$. Commuting the Hessian divergence gives $\mathrm{div}\,\mathrm{Hess}\, v = \mathrm{d}\Delta v + \mathrm{Ric}(\nabla v, \cdot)$; hence $\mathrm{div}\,\mathcal{Q}(v) = \mathrm{d}Lv = 0$. Since $r' = 0$ at $S$, $v_{ss} = 3v - \Delta_S v$. Subtracting $vB_{ss}$ gives $-\Delta_S v + (2 + 2m_0/a^3)v = \mathcal{D}v$, using $2m_0 = a + a^3$.

Let $\omega = N\,\mathrm{d}v - v\,\mathrm{d}N$. Product differentiation gives $\mathrm{sym}\,\nabla\omega = N\mathcal{Q}(v)$; mixed first derivatives cancel. Integrate against divergence-free $k$. At $S$, $\omega = -\kappa v\,\mathrm{d}s$, while the outward normal is $-\partial_s$, giving

$+\kappa \int_S v k_{ss}$. The end term is zero by hypothesis. Taking $k = \mathcal{Q}(v)$ gives $\int N|\mathcal{Q}(v)|^2 = \kappa \int_S v\mathcal{D}v$. Taking general $k$ gives the same cross term. Expanding the square proves (38). For a zero-boundary-trace TT difference the cross term vanishes. Since $N > 0$ in the interior, equality of weighted norms forces equality of tensors there and hence on the boundary by smoothness. □

This is a classical orthogonal-decomposition interpretation of the source square, not a new Penrose theorem.

A local extension keeps the defect for a symmetric tensor $k$ with the same boundary trace but nonzero divergence. Suppose $E_\infty = \lim_{R\to\infty} \int_{S_R} k(n_R, \omega)$ exists and the integrals converge. The same proof gives exactly

$$\int N|k|^2 - \kappa \int_S v\mathcal{D}v = \int N|k - \mathcal{Q}(v)|^2 + 2E_\infty - 2\int_M \langle \operatorname{div} k, \omega\rangle. \tag{39}$$

Indeed the cross term is now $\kappa \int_S v\mathcal{D}v + E_\infty - \int \langle \operatorname{div} k, \omega\rangle$. Substitution in the square proves the formula. If $E_\infty = 0$, weighted Cauchy–Schwarz yields

$$\left| \left( \int N|k|^2 - \kappa \int_S v\mathcal{D}v \right) - \int N|k - \mathcal{Q}(v)|^2 \right| \le 2 \left\| N^{-1/2} \operatorname{div} k \right\|_{L^2} \left\| N^{1/2}\omega \right\|_{L^2}. \tag{40}$$

Finiteness of the singularly weighted divergence norm at the horizon is a real assumption. Unweighted small divergence alone is insufficient. A nonzero trace prevents interpreting $k$ as a TT seed even though this integration identity still holds. No correction solver or finite-check guarantee is inferred.

## 11 Kernel, quartic order and radial equality

The source expresses

$$c_2(q) = \frac{1}{16\pi} \int N \left( |q - \mathcal{Q}(v)|^2 + |U|^2 + 2FJ_sF \right),$$

with sphere integration understood. Its complete kernel is

$$\mathcal{K} = \{\mathcal{Q}(v) : Lv = 0, \quad v|_S \in \operatorname{span}(1, x_1, x_2, x_3), \quad v \text{ decays}\}.$$

Trace-free spherical Hessians force degrees zero and one. Explicit degree-one CMC relations remove the remaining square. Continuous rotational averaging and closedness of a finite-dimensional kernel extend the finite-angular result, without a spectral gap on the complement.

The optional sign condition $q_{ss} \ge 0$ does not eliminate all dipole seeds. For $v|_S = c + d \cdot x$, it is $D_0 c \ge D_1 |d|$, with $D_0 = 2\kappa/a$, $D_1 = 6m_0/a^3$. Mixed radial and dipole modes may satisfy it.

For $q \in \mathcal{K}$, the quartic proof uses formal Gaussian-time Einstein jets modulo $\varepsilon^5$:

$$\partial_t g = 2K, \qquad \partial_t K = -\operatorname{Ric} -3g - (\operatorname{tr} K)K + 2K^2.$$

Bianchi propagation here is finite-jet constraint propagation, not an actual development theorem across the boundary. A slice $T = -\varepsilon v + \varepsilon^2 w$, with $Lw = 2\langle \mathrm{d}u, \mathrm{d}v\rangle - 12uv$ and zero boundary value, removes first-order extrinsic curvature. The new curvature starts at $\varepsilon^2 k$, with $k$ TT.

The end metric difference of the required $r^{-6}$ order preserves mass coefficients through degree four. Its second-order metric coefficient is

$$-2vq - Bv^2 - \mathrm{d}v \otimes \mathrm{d}v = -\operatorname{Hess}(v^2) + \mathrm{d}v \otimes \mathrm{d}v + Bv^2.$$

Null-cut area transport uses shear and expansion orders, not an assumption that every intermediate cut is minimal. The minimal-leaf height correction is $\varepsilon^2 z$, $z = \mathcal{D}^{-1} k_{ss}$, with area difference $\frac{1}{2}\varepsilon^4 \int z \mathcal{D} z$. This boundary term enters

$$c_3 = 0, \qquad c_4 = \frac{1}{16\pi} \int N \left( |k - \mathcal{Q}(v_k)|^2 + |U_2|^2 + 2F_2 J_s F_2 \right).$$

The source's geometric derivation of this identity remains an inherited dependency here.

Its zero-quartic analysis gives, for a nonradial component $V(s)x_i$,

$$(V' - (N/r)V)^2 = (6m_0/r^3)V^2.$$

For $z = V/r$, this is $|z'| = \sqrt{6m_0}\, r^{-3/2}|z|$. The coefficient is integrable at infinity and $z \to 0$. Gronwall backwards from $T$ gives

$$|z(s)| \le |z(T)| \exp\left( \int_s^T \sqrt{6m_0}\, r^{-3/2}\, \mathrm{d}s \right).$$

Sending $T \to \infty$ proves $z = 0$. Thus, conditional on the preceding coefficient identities, a nonradial kernel seed has $c_4 > 0$; the last elementary step does not replace those geometric identities.

### 11.1 Seed-dependent radii

**Proposition 11.1** (Local positivity with an actual remainder)**.** *For one fixed seed and branch, suppose $|\mathcal{D}_q - c_2\varepsilon^2| \le M_3\varepsilon^3$ on $0 \le \varepsilon \le e_0$, with $c_2 > 0$. Then $\mathcal{D}_q \ge (c_2/2)\varepsilon^2$ for $\varepsilon \le \min(e_0, c_2/(2M_3))$, with no second restriction if $M_3 = 0$. If $c_2 = c_3 = 0$, $c_4 > 0$, and $|\mathcal{D}_q - c_4\varepsilon^4| \le M_5\varepsilon^5$, the analogous radius is $\min(e_0, c_4/(2M_5))$.*

*Proof.* Subtract the absolute remainder bound, factor out the leading power of $\varepsilon$, and use the stated radius to bound the remaining factor below by half its positive leading coefficient. □

No numerical remainder constants or uniform branch radius are established here. As a seed approaches the kernel, $c_2$ may approach zero. Separate quadratic and quartic expansions do not produce a uniform neighborhood theorem without joint estimates. Finite coefficient samples do not fix this.

### 11.2 Exact radial equality, including turning points

For a radial TT seed

$$q = Cr^{-3}(\,\mathrm{d}s^2 - \tfrac{1}{2}r^2\sigma),$$

the source uses comparison uniqueness to make the actual small branch radial. Set $R = r\phi^2$, $\mathrm{d}\ell = \phi^2\,\mathrm{d}s$, $k = -\varepsilon C/(2R^3)$. Then

$$g = \mathrm{d}\ell^2 + R^2\sigma, \quad K = -2k\,\mathrm{d}\ell^2 + kR^2\sigma, \quad \mathcal{M} = \tfrac{R}{2}(1 + R^2 - \dot{R}^2 + R^2k^2).$$

Dots denote $\ell$-derivatives. The radial constraints are

$$\dot{k} = -3k\dot{R}/R, \qquad 2R\ddot{R} = 1 - \dot{R}^2 + 3R^2 - 3R^2k^2.$$

Differentiating directly gives

$$\dot{\mathcal{M}} = \frac{\dot{R}}{2}[1 + 3R^2 - \dot{R}^2 + 3R^2k^2 - 2R\ddot{R}] + R^3k\dot{k} = 0.$$

There is no division by $\dot{R}$; turning points cause no gap. The MOTS boundary has $\dot{R} = -Rk$, so $\mathcal{M}|_S = R(1 + R^2)/2 = b_*(4\pi R^2)$. At infinity $\phi = 1 + cr^{-3} + o(r^{-3})$ gives $p_0 = m_0 + 8c$, $p_i = 0$, and the same $\mathcal{M}$. The actual radial branch, not only its first four coefficients, therefore has zero deficit, conditional on the stated radial branch and asymptotics.

# 12 Benefits, costs and further work

## 12.1 Benefits specific to this composite

The full derivative preserves both the implicit change of variables $t = t(Z, \nabla f)$ and the prescribed boundary trace. It therefore provides an implementable local Newton operator for the actual equations, with a clear way to detect a missing coefficient, cutoff or conormal term. Its direction-free form removes an artificial zero-slope singularity. This is useful even when no inverse estimate is available: the object whose invertibility must be proved is now specified, rather than replaced by a convenient coercive scalar PDE.

The bordered face inverse retains the drift in the MOTS operator and uses the positive adjoint eigenfunction for compatibility. It explains which normalization and extra scalar are needed at a simple zero principal eigenvalue. Its perturbation bound identifies the actual missing quantitative datum, the norm of the unperturbed bordered inverse, instead of inferring stability from a determinant.

The period criterion separates total-charge conservation from conservation of all component fluxes and cohomology classes. It prevents a charged comparison that is adequate for a numerical mass bound from being overinterpreted as preservation of the original electromagnetic data. Likewise, the TT defect formula retains the inner orientation, the end pairing and the weighted divergence error. It can measure a particular approximate tensor's departure from the source identity, provided those integrals are controlled. The full-infimum metric estimate works without selecting or proving existence of a minimizer.

## 12.2 Costs and limitations

Exact differentiation is longer and contains $\nabla\alpha$, Hessian variations and state-dependent coefficient derivatives. A discretization must supply appropriate regularity and boundary lifts; changing coordinates does not eliminate these costs. The physical moment map may be poorly conditioned and dense. No benchmark here demonstrates a time, memory or accuracy advantage.

The relevant constants are not uniform for free. Axial ellipticity degenerates with slope; the face inverse may grow along a changing family; the AdS dipole gap tends to zero at infinity; and quadratic deficit coefficients vanish as seeds approach the four-dimensional kernel. Near the horizon the TT divergence bound has the singular weight $N^{-1}$. These are structural features, not floating-point inconveniences. Broad reading and the local checks do not remove the global geometric dependencies listed in the next section.

### 12.3 Concrete extensions and their obstacles

A first computational extension is to implement the displayed Jacobian on one fixed prepared truncation with conforming trial spaces, exact boundary lifts and physical mass matrices. Independent directional differences should include cutoff-transition states, beyond the constant-profile states tested here. Any claimed continuum root would additionally require quantified quadrature, truncation and off-space residual bounds, an actual bounded inverse on the stated spaces, and an invariant local ball. Increasing the finite matrix size alone proves none of these.

A second extension is a multiple-mode bordered face analysis along varying frontiers. One should track left and right kernels, their pairing with the chosen normalization functionals, and uniform Schauder or Sobolev estimates. Disconnected components require separate compatibilities; a higher-dimensional kernel requires a matrix of bordering functionals with a proved infimum singular value. Thus relaxing the simple-zero-mode hypothesis is possible in local Fredholm theory but needs data not supplied by the present proposition.

For charged data, preserving additional periods would require either restricting the discarded form to a zero-period class or retaining more fields with an explicit scalar-curvature budget. The source's one-field completion supplies no bound for that extra norm. Relaxing the source-free assumption would further introduce charge-density terms and change flux balances; it is not a harmless perturbation of the existing period proof.

For the AdS module, a useful next theorem would quantify the remainders jointly in distance to the kernel and seed size, with separate radial and dipole coordinates. It would have to reconcile quadratic and quartic scales and control the mass boundary functional. The present fixed-seed estimates do not furnish that joint bound. An approximate TT extension with weaker horizon assumptions would need a trace/Hardy estimate or a divergence correction respecting both the boundary trace and the end flux. Equation (40) alone does not prove either construction.

Finally, extending original-data rigidity to weaker regularity, different end classes or disconnected four-dimensional horizons requires revisiting the obstacle common-plane step, multiplier regularity, null-set cutoffs and static completion. These are specific obstacles to weakening the hypotheses. None is asserted solved by the local Ram reformulation.

## 13 Contribution and evidence limits

Applicability L4 is retained: full nonlinear differentiation, boundary-aware PDE realizations, physical finite coordinates and compatible flux forms directly match this source. It does not mean that four fifths of the proof has been verified.

The realized assessment is C2/5. Local contributions are the complete joint differential including cutoff and boundary terms (Proposition 5.1), the boundary-specific bordered inverse with its perturbation condition (Proposition 3.1), full-infimum and tied-envelope estimates (Propositions 6.1, 8.1), the charged period criterion (Proposition 7.1), and the weighted least-norm interpretation with exact defect (39). They are source-specific specializations and small extensions of classical arguments. Ram coordinates preserve the actual finite residual and physical norms; no new continuum inverse or verified speedup is demonstrated. A higher contribution rating would require a materially new estimate closing a global step, a further theorem, or demonstrated computational improvement with continuum error control. None is claimed.

The companion `M512-local-tests.py` checks the coefficient variations for $c = 4, 3$, including zero

slope; invariant quadratics against exact completions; the prescribed-trace boundary derivative; exact charged branch examples; metric-to-mass exponents; radial mass cancellation; and finite physical-coordinate identities. The test-result sidecar records 15 passing check groups, including 72 full-state directional-derivative comparisons and 200 comparisons of the two square completions. The maximum normalized derivative discrepancy was $3.86 \times 10^{-9}$, below the $2 \times 10^{-6}$ diagnostic tolerance. The full-state comparisons use the actual negative-$t$ profile where $p = n$, with zero slope included; separate symbolic checks retain the cutoff chain rule. Floating differences are diagnostics, not interval proofs. Symbolic checks certify only their algebraic expressions, not the PDE existence, trace or decay hypotheses needed geometrically.

The literal ledger records paths and intervals. All scientific source TeX files were read, but private corpus reading was explicitly partial. Hashes bind present bytes, not unchanged content throughout the reading interval: no pre-read hash was captured. The parent's dated cache receipt certifies current text identity, not stability throughout the earlier read interval. Public references distinguish bibliographic/abstract checks from whole-proof reading.

This complete scientific TeX was saved before testing and build metadata. Native editor/compiler dispatch, installed MiKTeX export, PDF text checks and visual inspection of final pages are recorded separately. Compilation is not mathematical certification. The final nested path/SHA256 freeze includes inputs and owned outputs, excludes its own recursive hash, and is checked by the existing manifest tool. Only M512-owned paths are writable.

The following remain unverified as whole arguments: rest-frame and strict-DEC reductions with all norm bookkeeping; trapped and obstacle frontier construction and regularity; global coupled existence with every parameter-uniform bound; limiting mass and area passages; the charged Riemannian endpoint; infinite-dimensional multiplier separation and regularization; arbitrary-null-set twist control; positive-mass rigidity at the stated regularity; global static classification and proper original-data embedding; and the complete AdS quartic geometric coefficient derivation. Their text and routes were read and analyzed. This report neither certifies them all nor announces a counterexample to them.

# A Literal read map

Paths are relative to the pinned composite's `build/`. The sidecar supplies individual exact intervals and hashes.

| Module | Literal full-text coverage |
|---|---|
| Neutral | Setup; 02 exterior reduction; 03 trapped domains; 04 deformation; 05 a priori; 06 elliptic existence; 07 mass limit. |
| Charged | All eight sections: introduction, preliminaries, rest, preparation, elliptic, comparison, mass and conclusion. |
| Rigidity, dimension three | Setup; full 08 variational rigidity and 09 static rigidity. |
| Dimension four | Setup; 01 end; 02 geometry; 03 system; 04 bounds; 05 regularity; 06 existence. |
| Rigidity, dimension four | 01 enclosures; 03 variation; 04 adjoint; 05 static base; 06 classification; 07 recovery. |
| Local AdS | All seven sections, 00 introduction through 06 conclusion, including full quadratic and quartic modules. |

| Module | Literal full-text coverage |
|---|---|
| Composite context | Introduction, charged extensions, neutral bridge, paper, preamble, modules and all three figure files. |
| Other source | Entire README and entire bibliography. |

Full reading means literal text exposure, not proof certification. Long files were split into consecutive chunks; exact intervals are retained in the machine-readable ledger.

Private proof paths, relative to their named corpus roots, are:

- `RamNonlin-IE-PDE-astra/TR8-lin-non-lin/ TR8-lin-non-lin.tex`, lines 1–320.
- `RamNonlin-IE-PDE-astra/TR1-lin-nonlin/stability.tex`, full file.
- `RamFoundationTRs/TR-rev-1-old-tr250/ TR-rev-1-old-tr250.tex`, lines 136–340 and 445–509.
- `res-rt/TechReps/FinalTechReps/TRs/ TR214-nonlinear-hammerstein-quasilinear-operators/ TR214-nonlinear-hammerstein-quasilinear-operators.tex`, lines 321–510.

The different Astra directory spellings are intentional. Confidential corpus paths are not public search terms. The partially read hypercircle discussion earns no credit.

## References


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[9] M. Subbarao, confidential local sources: Astra TR8; Astra TR1 stability supplement; revised foundation TR250; final TR214. Only literal ranges in Appendix A are invoked; these are not independently verified public citations.